DRAFT — P1–P5 MEASURED; P4-a-block-energy HIT (excess_final=0.08727119221832025); P5 2HIT/4MISS; L-cool-dir INSUFFICIENT

A3 quantitative prediction reference

Role: Compact MEASURED / FITTED / scored card for CS312 A3. MD only — **not** a PDF (PDF policy: report, intuition guide, optional quiz). Full A–G narrative lives in `docs/a3-predictive-intuition-guide.md` (Appendix Q).
Rule: Every number below is copied from store evidence. No invented coeffs or finals.
Sources: `media/p1/*`, `docs/a3-experimental-appendix.md`, intuition-guide Appendix Q1.1, `media/p2/batch2_partial_finals.csv` (jobs 0–6), `media/p2/p3_*_sharpness_summary.json`, `docs/a3-batch2-prediction-scorecard.md`, `internal/batch2-7of9-sgdr-score-status.md`, `internal/p1-scaling-fit-status.md`, `media/p4/a-adamw-{0,3,15,61,153,305,763,1526,3052,4578,6104,7630,9375}.summary.json` (+ `media/p4/stage-a/`), `internal/p4-lambda-early-fit.md`, `internal/p4-density-soft-0-1526.md`.

Labels: **MEASURED** = observed run/grid; **FITTED** = curve fit to MEASURED; **PREDICTION** = preregistered band; **HIT/MISS** = scored vs PREDICTION; **PENDING** = no Observed yet (Resolved blank).

P1 — Mode connectivity (MEASURED Batch 1)

[B = L(\theta_{\mathrm{avg}}) - \tfrac12\bigl(L(\theta_1)+L(\theta_2)\bigr),\quad \theta_{\mathrm{avg}}=\tfrac12(\theta_1+\theta_2).]

MEASURED summary (30-cell grid)

From `media/p1/barrier-summary.json` / `barrier-table.csv` / appendix heatmap:

Stat	Value
Cells	30
(B_{\min})	−0.1160578727722168 at ((x,y)=(250,100))
(B_{\max})	0.6237311363220215 at ((0,500))
(B_{\mathrm{mean}})	−0.033404584725697836
(B\le 0.01)	**26 / 30**
Failures ((B>0.01))	((0,250)), ((0,500)), ((50,500)), ((100,500))

Heatmap (rounded display; ✓ = (B\le 0.01)):

(x) \ (y)	25	50	100	250	500
0	−0.0288 ✓	−0.0416 ✓	−0.0275 ✓	0.2173	0.6237
50	−0.1009 ✓	−0.1053 ✓	−0.0910 ✓	0.0055 ✓	0.1299
100	−0.0932 ✓	−0.1019 ✓	−0.0972 ✓	−0.0512 ✓	0.0319
250	−0.0963 ✓	−0.1106 ✓	−0.1161 ✓	−0.0779 ✓	−0.0522 ✓
500	−0.0747 ✓	−0.0915 ✓	−0.0953 ✓	−0.0910 ✓	−0.0763 ✓
1000	−0.0597 ✓	−0.0769 ✓	−0.0855 ✓	−0.0855 ✓	−0.0825 ✓

Plots: `media/p1/plots/B_heatmap.png`, `B_vs_y_curves.png`, `B_vs_x_y500.png`, `ensemble_vs_avg.png`.

FITTED models A–C (Batch 1 MEASURED only)

Coefficients from intuition-guide **Appendix Q1.1** / `internal/p1-scaling-fit-status.md`. **Never label FITTED as MEASURED.**

Fit A — (B) vs (y) at (x=0)

Model	Domain	Equation	(R^2)	RMSE
Quadratic (preferred full curve)	(x=0), (y\in{25,50,100,250,500})	(B=-0.0765+7.61\times10^{-4},y+1.30\times10^{-6},y^2)	0.991	0.0243
Linear growth regime	(x=0), (y\ge 100) ((n=3))	(B=-0.1901+0.001628,y)	(\approx 1.000)	0.0003

Fit B — (B) vs (x) at (y=500)

Model	Domain	Equation	(R^2)	RMSE
Exp decay (**preferred**)	(y=500), six MEASURED (x)	(B=-0.0663+0.6870,e^{-0.02316,x})	0.995	0.0174

(Weaker candidates in Q1.1: (\log(1+x)), (a+b/\sqrt{x+1}) — not preferred.)

Fit C — ensemble–avg gap (G) vs (B)

[G=L(\theta_{\mathrm{avg}})-L(p_{\mathrm{ens}}).]

Model	Domain	Equation	(R^2)	RMSE
Linear (all cells)	30 MEASURED cells	(G=0.0897+1.0328,B)	0.989	0.0151
Linear ((B>0.01))	4 failure cells	(G=0.1020+1.0066,B)	0.9997	0.0039

Out-of-grid intervals (`P1-B-x0-y750`, …) are Kind=`extrapolated` **PREDICTION** — not Observed (see ledger / Q1.1).
P1(d) cosine (\Delta B) (`P1-d-dB-cos`) — **PENDING** (no cosine-fork MEASURED).

P2 — Schedule ranking (MEASURED @ Batch 2 9/9)

Source: `media/p2/batch2_partial_finals.csv` jobs 0–3; scorecard `docs/a3-batch2-prediction-scorecard.md`.
Best raw final (L^\star=2.9282028675079346) (`linear`). Regret (R=L-L^\star).

Rank	Arm	W&B	Final `val_loss` (exact)	(R) vs linear
1	`linear`	`r0294p93`	2.9282028675079346	0
2	`cos`	`sktw67q8`	2.934119462966919	0.005916595458984375
3	`wsd0.2`	`soosjcxr`	2.93774151802063	0.009538650512695312
4	`wsd0.0`	`widt420l`	3.0975050926208496	0.16930222511291504

Ranking (lower val loss better):
(\mathrm{linear}\prec\mathrm{cos}\prec\mathrm{wsd0.2}\prec\mathrm{wsd0.0}).

`P2-sched-rank`: (\min(L_{\mathrm{cos}},L_{\mathrm{wsd0.2}},L_{\mathrm{linear}}) < L_{\mathrm{wsd0.0}}) — **HIT**.

P2 scored counterfactuals (jobs 4–6 — MEASURED)

SGDR final (job 6): `a3-b2-p2-sgdr2000` W&B `51b3qq4n`, `val_loss=3.0348048210144043`. Tuned baseline = `linear`.

ID	Quantity	PREDICTION band	Observed (exact)	Score
`B2-P2-dip-R` / `P2-dip-final-R`	(R=L_{\mathrm{dip}}-L_{\mathrm{wsd0.2}})	([-0.02,0.02))	(0.0012464523315429688) (= (2.938987970352173-2.93774151802063))	**HIT**
`B2-P2-invsqrt-dL`	(\Delta L=L_{\mathrm{invsqrt}}-L_{\mathrm{linear}})	([0.01,0.08))	(0.09225320816040039) (= (3.020456075668335-2.9282028675079346))	**MISS** (falls in ([0.08,\infty)))
`B2-P2-sgdr-dL`	(\Delta L=L_{\mathrm{sgdr}}-L_{\mathrm{linear}})	([0.01,0.08))	(0.10660195350646973) (= (3.0348048210144043-2.9282028675079346))	**MISS** (falls in ([0.08,\infty)))
`B2-P2-sgdr-dL` (vs invsqrt)	(\lvert L_{\mathrm{sgdr}}-L_{\mathrm{invsqrt}} \rvert)	([0,0.03))	(0.014348745346069336) (= (\lvert 3.0348048210144043-3.020456075668335\rvert))	**HIT**

P2-horizon-dL (both HF at full stop): invsqrt + SGDR (\Delta L) both **MISS**; |SGDR−inv| **HIT**. Mid-dip (R_{\mathrm{mid}}) — **PENDING** (not in partial CSV). EMA gap (P2-ema-gap-frac) — **PENDING**.

P3 — Edge of stability (MEASURED starters; η-change PENDING)

Sources: media/p2/p3_sgd_sharpness_summary.json, media/p2/p3_adamw_sharpness_summary.json.

ID	Quantity	PREDICTION band	Observed (exact)	Score
B2-P3-sgd-lam / P3-clean-edge	Median (\lambda_{\max}) updates 1200–1500	([150,250))	(245.16314737956807)	**HIT**
B2-P3-adamw-pre / P3-mb-edge	Post-warmup median (\eta\lambda^{\mathrm{pre}})	([30,50))	(30.987715914514)	**HIT**
P3-eta-half / P3-eta-double	Mid-run (\eta) ×0.5 / ×2	see ledger	—	**PENDING** (not in B2 required 9)
P3(c) hparam exploration	—	—	—	**PENDING**

Handout thresholds (mechanism, not Observed): GD edge (\eta\lambda=2) ⇒ (\lambda=200) at (\eta=0.01); Adam (\beta_1=0.9) precond edge (38).

P4 — Hessian trajectory (MEASURED stage a 13/13)

Sources: media/p4/stage-a/a-adamw-{0,3,15,61,153,305,763,1526,3052,4578,6104,7630,9375}.json (+ summaries). Stage a **DONE 13/13**.

MEASURED card ((\lambda_{\max}))

Job	Step	(\lambda_{\max}) (exact)	Status
a-adamw-0	0	1431.5030691763686	**MEASURED**
a-adamw-3	3	1571.7420262270407	**MEASURED**
a-adamw-15	15	53.36426571742344	**MEASURED**
a-adamw-61	61	18.054717868295125	**MEASURED**
a-adamw-153	153	6.502395300121604	**MEASURED**
a-adamw-305	305	4.32926630998023	**MEASURED**
a-adamw-763	763	3.02398152405625	**MEASURED**
a-adamw-1526	1526	2.1954286466134665	**MEASURED**
a-adamw-3052	3052	3.1637529312879873	**MEASURED**
a-adamw-4578	4578	3.7423705126158477	**MEASURED**
a-adamw-6104	6104	4.155164815834522	**MEASURED**
a-adamw-7630	7630	5.281516516770698	**MEASURED**
a-adamw-9375	9375	5.967310256347902	**MEASURED**

Trajectory: collapse by 15 → minimum @1526 ((2.195)) → rebound (2.20\to 3.16\to 3.74\to 4.16\to 5.28\to\mathbf{5.9673}) through **9375** (exact (5.967310256347902); **inside** handout `[4.5,6.0]`). Preferred early-fit asymptote ≈(3.51) still **undershoots**. See `internal/p4-lambda-rebound-13of13.md`.

Density (file-exact → ledger): neg_frac 0→153→1526→9375 = (0.004510114309467541\to 0.01491374251600709\to 0.9344596369432798\to 0.06965100420936976). Final in `[0,0.15)` but **did not shrink** vs init → `P4-a-density-neg` **MISS**.

Early-fit scorecard

ID	Band	Preferred point	Observed	Score
`P4-lam-1526-earlyfit`	`[1.4,4.0)`	≈3.51	**2.1954286466134665**	**HIT** (preferred note-only miss)
`P4-lam-9375-earlyfit`	`[0.3,4.0)`	≈3.51	**5.967310256347902**	**MISS** (preferred also miss; inside handout `[4.5,6.0]`)

P4-a scorecard (from `internal/p4-score-dryrun.md` → ledger)

ID	PREDICT (locked)	Score	Observed (file-exact)	Predictive mechanism note
`P4-a-family-sharpest`	late attn|mlp; ratio `[1.5,5)`	**MISS**	late=`head` ((5.641603135007765)); early_top2=`embedding,attention`; ratio=(7.38436579244244)	Pre-reg: residual-block late sharpening. Fail: head stayed sharpest @9375.
`P4-a-norm-sharpness`	mid→final nondec.; final λ in `[4.5,6.0]`	**HIT**	(\lambda) (2.1954286466134665\to5.967310256347902); tr nondecreasing; final in `[4.5,6.0]`	Edge-band + mid→final sharpening held.
`P4-a-density-neg`	shrink 0→9375; final `[0,0.15)`	**MISS**	(0.004510114309467541\to\cdots\to0.06965100420936976); shrink_0→9375=false; final band OK	Pre-reg: saddle cleanup. Fail: final neg_frac > init.
`P4-lam-9375-earlyfit`	λ@9375 ∈`[0.3,4.0)`	**MISS**	(5.967310256347902) (inside handout `[4.5,6.0]`)	Pre-reg: early-decay / asymptote (\sim 3.51). Fail: rebound into handout edge.
`P4-a-block-energy`	family>random; excess `[0.05,0.30)`	**HIT**	family_within-random excess_final=0.08727119221832025; fam_above=True; layer_above=True	MEASURED `a-random-energy-1526/9375`.

Operator copied DRAFT_* into ledger (script `--apply` does not mutate docs). Summary: `internal/p4-9375-hit-miss-summary.md`. PREDICT bands unchanged.

P4(b) soft baseline (mirrors report — not an invar score)

Field	Value
Source	`media/p4/b-rescale-reference.json`
Status	**MEASURED** baseline only (`scale={}`)
(\lambda_{\max})	(5.967310256347902) (≈5.967; matches stage-a @9375)
`probe_loss`	(2.9474442899227142)
Hand-rescale arms	`b-rescale-head-x0.25` / `x0.5` / `x2` — **launched / waiting** (`ap-lSeukZDR7yWisgODr61i1k`)
`P4-invar-dL`	**MISS** —
Dry-run scorer	`internal/p4-score-invar-from-media.py` (exit 2 until after-arm)
Waiting note	`internal/p4-b-waiting.md`

PREDICT (locked, unscored): (\lvert\Delta L\rvert\in[0,0.002)); λ ratio (\in[2,\infty)\cup(0,0.5)).

P4 b/c scorecard (from dry-run → deliverables)

ID	PREDICT (locked)	Score	Observed (file-exact)
`P4-top10-frac`	mean frac ∈[0.6,0.85)	**MISS**	mean top_frac=0.09296940652964016 (source=c-full)
`P4-c-loss-rank`	full≈top<removed; gap ∈[0.02,0.20)	**MISS**	L_full=3.582527816295624; L_top=5.502330541610718; L_removed=3.5821831226348877; gap=-0.000344693660736084

P5 — Cooldown barrier (MEASURED from dry-run)

Source: `media/p5/` / `internal/p5-score-dryrun.md`.

ID	PREDICT (locked)	Score	Observed (file-exact)
`P5-B-cool-x0`	B_C ∈ (-∞,0.01]	**MISS**	B_C=0.7697049379348755
`P5-dB-cool`	ΔB ∈ [0.5,0.8)	**MISS**	dB=-0.14600493793487546 (B_R=0.6237 [JSON] – B_C=0.7697049379348755)
`P5-B-const-ctrl`	B_K ∈ [0.5,∞)	**HIT**	B_K=0.7320129871368408
`P5-rank-arms`	B_C < B_R and B_C < B_K	**MISS**	B_C=0.7697049379348755 B_R=0.6237 B_K=0.7320129871368408
`P5-G-cool-fitC`	if B_C≤0.01: G∈[0,0.12); else	G-(B+0.09)	<0.05
`P5-L-cool-dir`	L_C<L_R; ΔL ∈ [0.02,0.15)	**INSUFFICIENT**	dL / (L_R, L_C) missing (`media/p5/` dL=null) — no HIT/MISS
`P5-emergent-basin`	same-valley (B_C≤0.01 and B_K>0.01)	**MISS**	B_C=0.7697049379348755 B_K=0.7320129871368408 → not-same-valley (JSON emergent='true-modes')

PENDING (no Observed — do not invent)

Topic	Status	Notes
Mid-dip (R_{\mathrm{mid}})	**PENDING**	Not in `batch2_partial_finals.csv`
EMA gap (`P2-ema-gap-frac`)	**PENDING**	No scored EMA series
P3 η-change / P3(c)	**PENDING**	Not in required 9-job finals
P4 `P4-a-block-energy`	**HIT**	excess_final=0.08727119221832025; fam_above=True; layer_above=True
P5 `P5-L-cool-dir`	**INSUFFICIENT**	dL=null / (L_R, L_C) missing — no invent HIT/MISS

Quiz answer key: Resolved filled only for MEASURED / FITTED / EXTRAPOLATED rows; `P4-a-block-energy` **HIT** (Resolved filled); `P5-L-cool-dir` stays Resolved = — · PENDING/ INSUFFICIENT (**P5** otherwise MEASURED 6/7).

Cross-links

Resource	Path
Full A–G + Appendix Q	`docs/a3-predictive-intuition-guide.md`
Practice quiz + key	`docs/a3-counterfactual-prediction-quiz.md`
Experimental tables	`docs/a3-experimental-appendix.md`
Batch 2 scorecard	`docs/a3-batch2-prediction-scorecard.md`
Ledger	`docs/a3-prediction-ledger.md` / `.csv`
Fit status	`internal/p1-scaling-fit-status.md`
P4 early λ fit	`internal/p4-lambda-early-fit.md`
P1 media	`media/p1/`
P2 finals CSV	`media/p2/batch2_partial_finals.csv`
P3 sharpness JSON	`media/p2/p3_*_sharpness_summary.json`